

Deep Snake on FLAME-1: Detector-Coupled vs. Detector-Free

Re-running the deep-contour methods on a visible-flame dataset

CS269 Project — Deformable Models

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1 Summary

We re-ran the two deep-contour “snake” methods on **FLAME-1** (Sycan Marsh, 2,001 RGB frames with hand-labeled *visible-flame* PNG masks; contiguous 70/15/15 split $\rightarrow$ 1,400 train / 300 val / 301 test). Two implementations were compared:

- **deep_snake_simple** — the original detector-free design: a shared U-Net produces a coarse mask whose connected components seed the contour, which is then refined by circular-convolution snake heads.
- **deep_snake_paper** — a paper-faithful re-implementation that *decouples detection from contour refinement*: a separately-trained MMDet **CenterNet** detector localizes fire boxes, an octagon is initialized from each box, and the snake deforms it. The snake is trained on ground-truth boxes; at test time it consumes the detector’s predicted boxes.

Table 1: Test results on FLAME-1 (301 frames, IoU/Dice on binary masks). Detector trained 10 epochs; snakes 30 epochs.

Method	IoU	Dice	Frames with a prediction
deep_snake_simple (detector-free)	0.658	0.792	301 / 301 (100%)
deep_snake_paper (CenterNet $\rightarrow$ snake)	0.050	0.080	73 / 301 (24%)

2 Headline finding: the task, not the architecture

The single most important result is the contrast with the recorded **FLAME-3** numbers. On FLAME-3 (where ground truth is the paired thermal frame thresholded at $\geq 150^\circ\text{C}$ —a smoke-occluded hot region that is essentially *invisible* in RGB), the deep snake scored IoU 0.032, below even the learned U-Net (0.147) and DALs (0.122), and far below oracle-initialized GAC (0.365). The color floor was 0.004.

On FLAME-1, where the label *is* the orange flame visible in RGB, **deep_snake_simple** jumps to **IoU 0.658 / Dice 0.792**—a $\sim 20\times$ improvement, and higher than *any* FLAME-3 method including the oracle-seeded classical contours, with *no* oracle prior. This is direct evidence that Deep Snake’s weak FLAME-3 showing was a property of the *task* (RGB cannot see thermal-defined fire), not of the method.

Table 2: FLAME-3 (recorded, RESULTS.md) vs. FLAME-1 (this run).

	FLAME-3 IoU	FLAME-1 IoU
Color floor (R–G)	0.004	—
U-Net (RGB)	0.147	—
DALS (RGB)	0.122	—
Oracle-init GAC	0.365	—
Deep Snake (detector-free)	0.032	0.658

3 Why the paper-faithful pipeline underperforms here

`deep_snake_paper` scored only 0.050 overall—but this is a **detector-recall bottleneck**, not a snake failure:

- The CenterNet detector reached **mAP 0.120 / mAP@50 0.430** after 10 epochs. Notably, this is a *working* detector: on FLAME-3 the same pipeline trained to mAP 0.000 (it never learned), and was unusable. The fix that made it train at all was correcting the learning rate (the base config’s LR of 0.01 is calibrated for an effective batch of 128; at our batch of 4 the heatmap head collapsed—auto-scaling to $\approx 3 \times 10^{-4}$ resolved it).
- At test time the detector fires on only **24% of frames**. Where it misses, the snake receives no box and produces an empty mask, dragging the mean to 0.050.
- *Conditioned on the detector firing*, the snake reaches IoU **0.206**—over $4\times$ the unconditional score. Its oracle-box validation IoU was 0.180. So the contour refinement works; the pipeline is gated by how often the detector finds the fire.

The decoupled design pays for an explicit detection stage that, at 10 epochs on small/fragmented fire boxes (median ~ 2 dominant components per frame), has low recall. The detector-free variant sidesteps this by seeding from a dense coarse mask, which on a visible-flame dataset is far more reliable.

4 Takeaways

1. **Deep Snake is competitive when the target is visible in RGB.** The detector-free snake hits IoU 0.658 on FLAME-1 vs. 0.032 on FLAME-3—the dataset, not the model, drove the original poor result.
2. **The paper-faithful detector→snake pipeline is recall-limited.** The snake itself refines well (0.206 IoU when given a box, 0.180 oracle), but a 10-epoch CenterNet only localizes 24% of test frames. More detector training (epochs were capped for runtime; it was still improving) and less-fragmented box targets are the obvious levers.
3. **The pipeline is now functional end-to-end**, which it was not on FLAME-3 (detector mAP 0). The remaining gap is detector capacity/training, a tractable next step.

Artifacts: `results/rerun_flame1_20260601_193349/` (per-frame CSVs, logs, detector checkpoint, provenance). Detector: 10 epochs; snakes: 30 epochs; RTX 4090. Recorded additively; committed FLAME-3 results and `RESULTS.md` unchanged.